

10

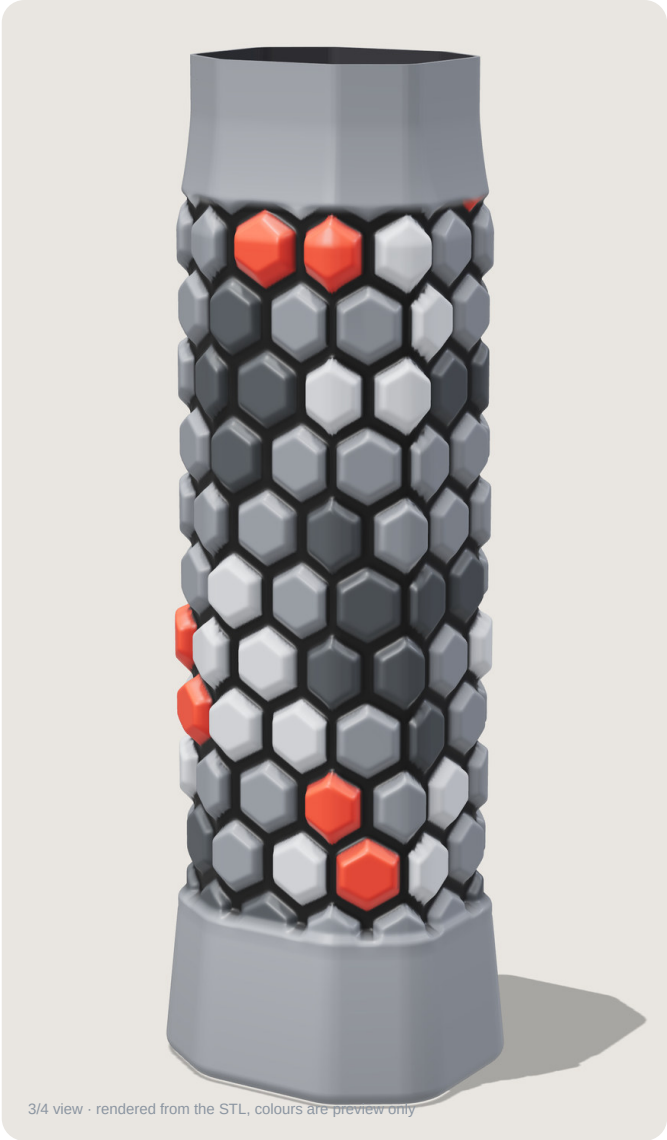
# HEXA-MOD

Modular hex pads on four height levels, colour-coded (red = tallest).

47.0 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

|                          |                               |
|--------------------------|-------------------------------|
| Overall height           | 132.82 mm                     |
| Max footprint (foot)     | 42.63 × 36.61 mm              |
| Bore                     | 32.06 × 26.04 mm, 127.8 deep  |
| Textured band            | z 21.5 – 116.0 mm (94.5 tall) |
| Outer size in band (max) | 38.7 × 33.0 mm                |
| Peak texture height      | 3.50 mm above bore            |
| Base wall / min rim      | 1.2 mm / 0.8 mm               |
| Volume / surface area    | 37.92 cm³ / 325 cm²           |

WEIGHT · 100 % INFILL

|      |            |                        |        |
|------|------------|------------------------|--------|
| PLA  | 1.24 g/cm³ | <div><div></div></div> | 47.0 g |
| PETG | 1.27 g/cm³ | <div><div></div></div> | 48.2 g |
| TPU  | 1.21 g/cm³ | <div><div></div></div> | 45.9 g |
| ABS  | 1.04 g/cm³ | <div><div></div></div> | 39.4 g |

50 g limit

COLOURWAY

|             |             |             |             |             |                   |
|-------------|-------------|-------------|-------------|-------------|-------------------|
| <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 3MF print palette |
| #1C1C1E     | #464A4E     | #74787E     | #B0B4BA     | #D63A28     |                   |

PRINT PROFILE

|                       |                        |
|-----------------------|------------------------|
| Infill / layer height | 100 % / 0.20 mm        |
| Layers                | 665                    |
| Orientation           | upright, butt end down |
| Supports              | none required          |
| Filament, 1.75 mm     | ≈ 15.8 m (100 % flow)  |

DESIGN INTENT

A honeycomb of individual hex pads, each separated by a 1.7 mm gap and set at one of four heights. The height map is pseudo-random with a gentle swell along the handle; the tallest pads (red) reach the full 3.5 mm crest.

*Pads are chamfered on the lower edge so they never overhang beyond 45 degrees.*

VERIFIED AGAINST THE REFERENCE

|                             |                                      |
|-----------------------------|--------------------------------------|
| Bore vs reference           | identical                            |
| Outer envelope vs reference | +0.001 mm at most                    |
| Rim vs reference            | +0.40 mm (deliberate, ≥ 0.8 mm wall) |
| Bodies / enclosed voids     | 1 / none                             |
| Material under 0.4 mm wide  | 0.000 % of volume                    |
| Longest bridge              | none                                 |

KEY FEATURES

- 12 columns × 11 rows of pads, 10.2 mm across flats
- Four levels: 2.35 / 2.76 / 3.13 / 3.50 mm above the bore
- Red pads mark the tallest level (about 6% of the band)

FILES

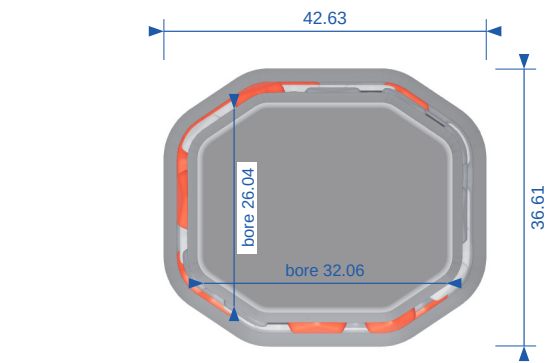
|                     |                             |
|---------------------|-----------------------------|
| stl/10_hexa_mod.stl | 16.2 MB · 324,380 triangles |
| 3mf/10_hexa_mod.3mf | 3.8 MB · colour             |

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

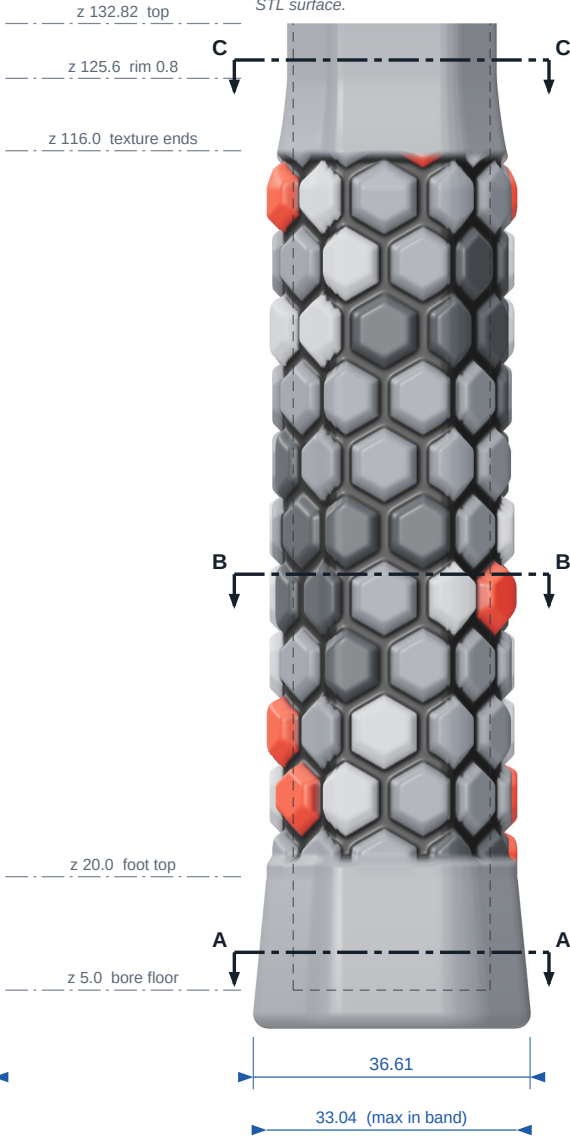
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

|                                  |               |
|----------------------------------|---------------|
| Overall height                   | 132.82        |
| Overall width × depth (foot)     | 42.63 × 36.61 |
| Bore width × depth               | 32.06 × 26.04 |
| Bore depth (floor at z 5.0)      | 127.82        |
| Outer W × D, textured band (max) | 38.70 × 33.04 |
| Foot top (z 20)                  | 39.16 × 33.13 |
| Butt face (z 0)                  | 40.56 × 34.54 |
| Peak texture height above bore   | 3.50          |
| Base wall / minimum rim wall     | 1.20 / 0.80   |

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

HEXA-MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip\_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

UNITS

millimetres

PROJECTION



MATERIAL

PLA / PETG / TPU

INFILL

100 %

LAYER HEIGHT

0.20 mm

MASS (PLA)

47.0 g

VOLUME

37.92 cm³

SURFACE AREA

325 cm²

REVISION

2

DATE

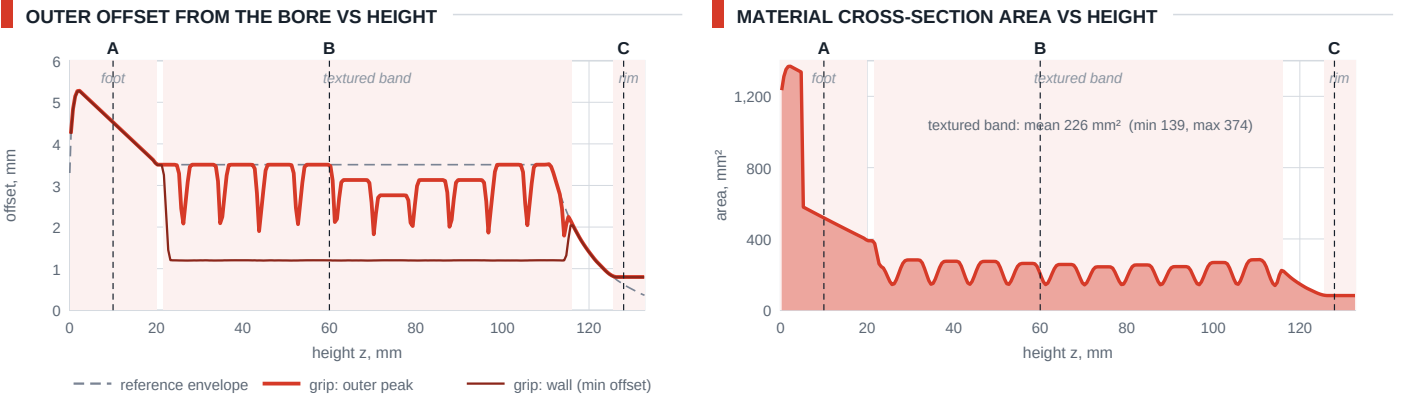
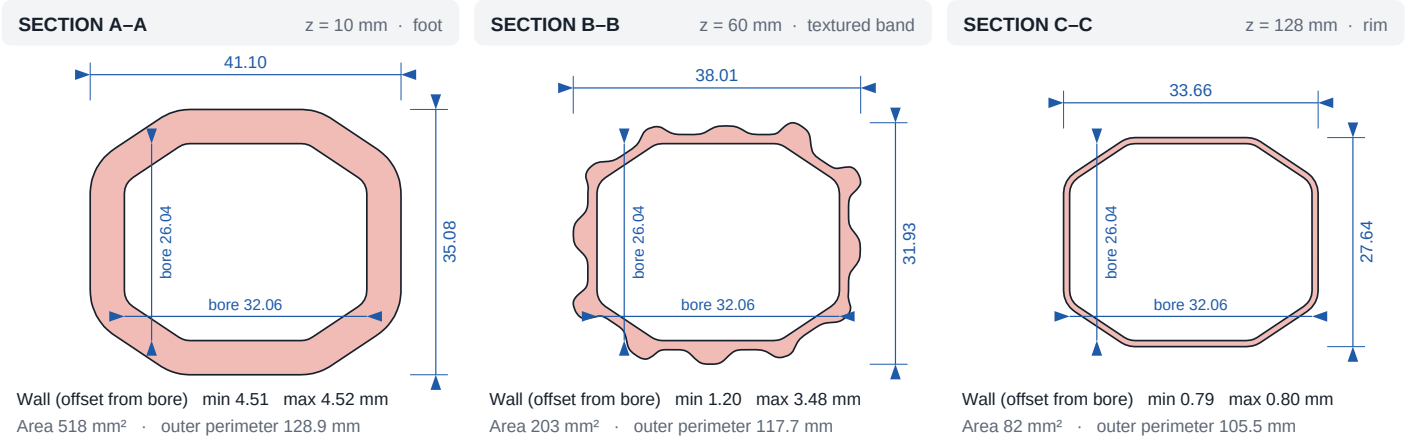
2026-09-30

SHEET

2 of 4

# Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

|                             |                                               |
|-----------------------------|-----------------------------------------------|
| Overall length (Z)          | 132.82 mm                                     |
| Overall width (X)           | 42.63 mm                                      |
| Overall depth (Y)           | 36.61 mm                                      |
| Bore width × depth          | 32.06 × 26.04 mm                              |
| Bore area / perimeter       | 751.0 mm <sup>2</sup> / 100.5 mm              |
| Bore depth (floor at z 5.0) | 127.82 mm                                     |
| Butt face (z 0)             | 40.56 × 34.54 mm                              |
| Outer W × D in band (max)   | 38.70 × 33.04 mm                              |
| Circumference in band       | 110.3 – 119.6 mm                              |
| Volume / surface area       | 37.92 cm <sup>3</sup> / 325.3 cm <sup>2</sup> |

ZONES BY HEIGHT

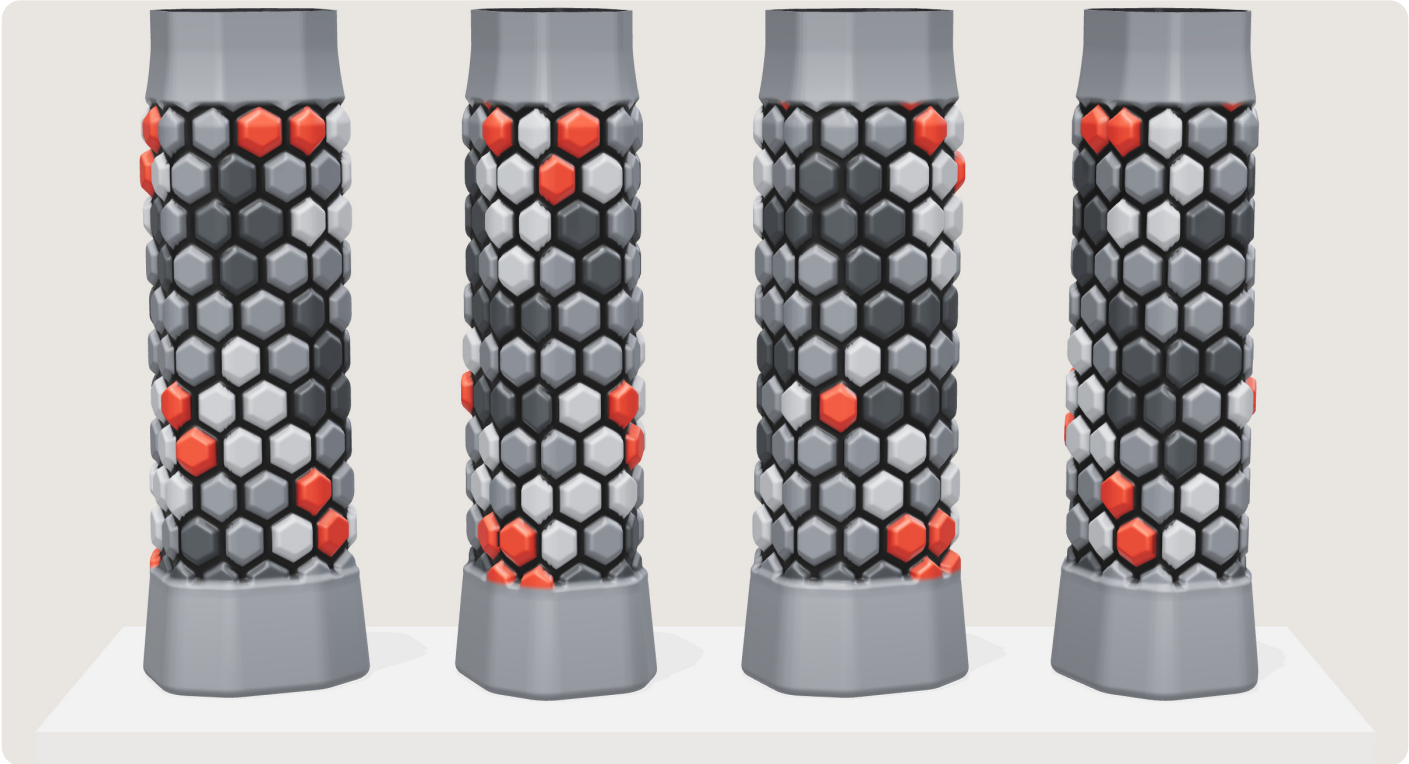
| ZONE                              | Z FROM | Z TO   | HEIGHT         |
|-----------------------------------|--------|--------|----------------|
| Solid floor (butt)                | 0.0    | 5.0    | 5.0            |
| Flared foot                       | 0.0    | 20.0   | 20.0           |
| Smooth collar                     | 20.0   | 21.5   | 1.5            |
| Textured band                     | 21.5   | 116.0  | 94.5           |
| Smooth neck (taper)               | 116.0  | 125.6  | 9.6            |
| Rim (0.8 mm wall)                 | 125.6  | 132.82 | 7.3            |
| Base wall (design / measured min) |        |        | 1.20 / 1.19 mm |
| Rim wall (design / measured min)  |        |        | 0.80 / 0.79 mm |
| Peak texture height above bore    |        |        | 3.50 mm        |

OUTER DIMENSIONS BY HEIGHT

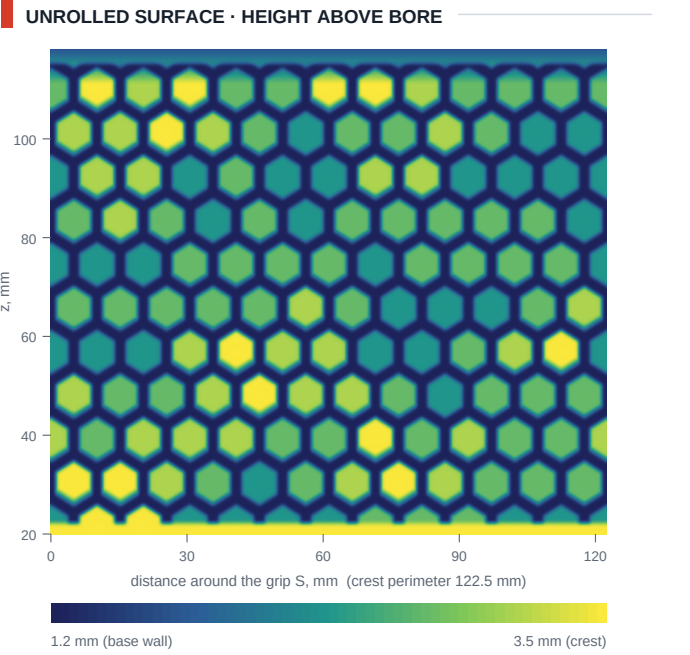
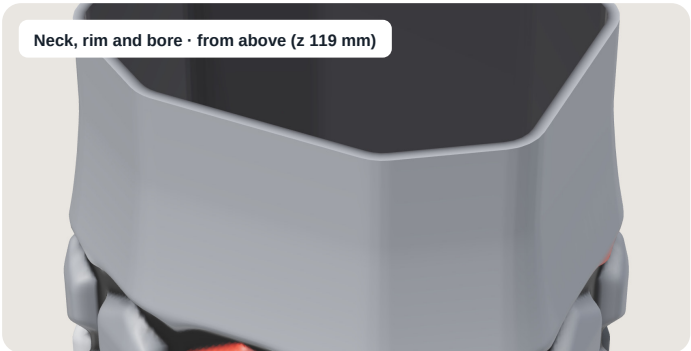
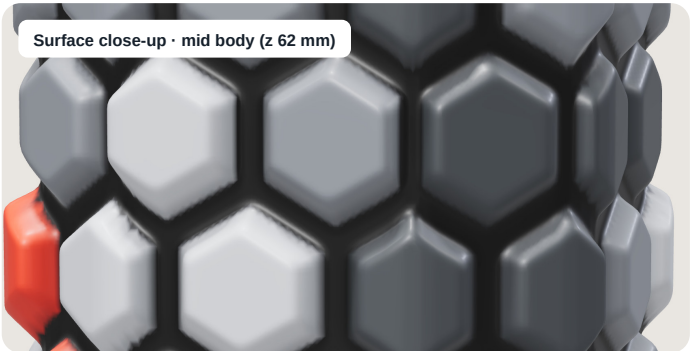
| Z (MM) | OUTER W | OUTER D | PERIMETER | AREA MM <sup>2</sup> | WALL MIN | PEAK OFFSET |
|--------|---------|---------|-----------|----------------------|----------|-------------|
| 5.0    | 42.08   | 36.06   | 132.0     | 958                  | 5.00     | 5.01        |
| 10.0   | 41.10   | 35.08   | 128.9     | 518                  | 4.51     | 4.52        |
| 20.0   | 39.11   | 33.09   | 122.6     | 393                  | 3.52     | 3.52        |
| 30.0   | 38.33   | 32.67   | 118.8     | 282                  | 1.20     | 3.50        |
| 45.0   | 37.80   | 32.03   | 115.1     | 207                  | 1.20     | 3.13        |
| 60.0   | 38.07   | 31.91   | 117.6     | 203                  | 1.20     | 3.48        |
| 75.0   | 37.59   | 31.49   | 117.0     | 244                  | 1.20     | 2.76        |
| 90.0   | 37.15   | 32.20   | 117.7     | 229                  | 1.20     | 3.13        |
| 105.0  | 37.70   | 31.09   | 112.9     | 163                  | 1.20     | 2.94        |
| 116.0  | 36.26   | 30.24   | 113.4     | 221                  | 2.04     | 2.10        |
| 125.0  | 33.77   | 27.75   | 105.8     | 88                   | 0.85     | 0.85        |
| 132.0  | 33.66   | 27.64   | 105.5     | 82                   | 0.79     | 0.80        |

# Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only



| DESIGN PARAMETERS             |                              |
|-------------------------------|------------------------------|
| Pad grid                      | 12 × 11 (pointy-top hex)     |
| Pad across flats              | 10.21 mm                     |
| Gap between pads              | 1.7 mm                       |
| Levels                        | 2.35 / 2.76 / 3.13 / 3.50 mm |
| Gap floor                     | 1.20 mm (base wall)          |
| Pad edge                      | chamfered, about 0.8 mm      |
| Level colours                 | 3 greys + red                |
| PRINT & VERIFICATION          |                              |
| Result: fitment / print-ready | PASS / PASS                  |
| Bodies · genus                | 1 · 0                        |
| Bore vs reference             | identical                    |
| Envelope over reference       | +0.001 mm                    |
| Material under 0.4 mm wide    | 0.000 %                      |
| Longest bridge                | none                         |